

Target Detection Based on Dual-Domain Sparse Reconstruction Saliency in SAR Images

Lu Li , Lan Du , Senior Member, IEEE, and Zhaocheng Wang 

Abstract—The conventional target detection methods in synthetic aperture radar (SAR) images usually utilize the clutter information via the clutter statistical modeling. However, it is difficult to select an appropriate clutter statistical distribution model for a real SAR image. To make full use of the clutter information while avoid the clutter statistical modeling, in this paper, we propose a novel target detection method based on the dual-domain sparse reconstruction saliency in SAR images. Considering the speckle noise in SAR images, we utilize not only the intensity information in image domain but also the structure information in feature domain. First, the image-domain saliency map is constructed via the sparse reconstruction on clutter templates extracted from pure clutter SAR images in the image domain with the pixel-level intensity information as the descriptor, and the potential target templates are obtained from this saliency map. Then, the feature-domain saliency map is constructed via the sparse reconstruction on both clutter and potential target templates in the feature domain with the superpixel-level locality-constrained linear coding (LLC) of pixel-level scale invariant feature transform features as the descriptor, where the LLC can describe the spatial and structural information in superpixel level. Finally, the aforementioned two saliency maps are integrated to make our method more robust to speckle noise, and then, the winner-take-all approach is applied to detect targets from the integrated saliency map. We evaluate the proposed method on the miniSAR data and the RADARSAT-2 data. On the miniSAR data, the proposed method is at least 5% higher in terms of AUC than the other saliency methods (i.e., the Itti's method, the SR method, the SVD method, and the JDSR method), and at least 20% higher in F1-score, 22% higher in figure of merit than the Gaussian constant false alarm rate (CFAR), the Gamma-CFAR, and the aforementioned saliency methods. On the RADARSAT-2 data, the proposed method also achieves better performance than the previously compared methods.

Index Terms—Dual domain, synthetic aperture radar (SAR), sparse reconstruction, saliency, target detection.

I. INTRODUCTION

As an active microwave imaging system, synthetic aperture radar (SAR) is a hot topic in the field of radar research. Compared with optical monitors, the well-known advantage of

the SAR imaging is that it can provide remote sensing data under all-day and all-weather conditions. Furthermore, with the development of the SAR sensor technology, the resolution of SAR images becomes higher and higher. More and more SAR images have been widely applied in various fields. With the overwhelming amount of SAR images available, the demand of the SAR image interpretation is gradually increasing.

In the procedure of SAR image interpretation, target detection, which is the first stage in the SAR automatic target recognition, is becoming increasingly desirable. Up to now, there are many target detection methods available for SAR images in literatures. Among these methods, due to the simple computation and the adaptive threshold, the constant false alarm rate (CFAR) methods [1], [2], which are based on the clutter statistical distribution model, are usually effective and have been extensively studied for target detection in SAR images. One widely used CFAR detector is the two-parameter CFAR detector [1] that assumes the background clutter obeys the Gaussian statistical distribution. Lopez *et al.* [2] proposed a Gamma-based CFAR detector, which regards the Gamma distribution as the clutter statistical distribution model. Although these aforementioned CFAR methods utilize the clutter information in target detection through clutter statistical modeling, it is difficult to select an appropriate clutter statistical distribution model especially in complex scenes and an inappropriate clutter statistical distribution model will lead to poor detection performance. Therefore, what mentioned previously inspires us to develop an effective target detection method for SAR images without the clutter statistical modeling.

In the past couple of decades, saliency detection has received a great deal of attention in the field of computer vision, such as image segmentation [3], object detection [4], [5], recognition [6], etc. Aiming at detecting and locating certain regions or objects to which one's attention can be attracted in a scene, saliency detection can allocate more computational resource for rapid scene understanding. Lots of image saliency detection methods have been proposed and different methods mainly focus saliency detection on different saliency cues.

Taking the center-surround difference as saliency cue, Itti *et al.* [7] used Gaussian pyramids to compute center-surround difference based on multiscale image features, including colors, orientations, and intensity. Liu *et al.* [8] proposed a visual attention detection method that combines Itti's pyramid model with the singular value decomposition (SVD). Wang *et al.* [9] modified Itti's saliency model through introducing target's prior size information to select some task-dependent scales for

Manuscript received January 24, 2018; revised April 24, 2018 and August 26, 2018; accepted September 18, 2018. Date of publication October 18, 2018; date of current version November 27, 2018. This work was supported in part by the National Science Foundation of China under Grant 61771362, Grant U1833203, and Grant 61671354, and in part by the 111 Project (B18039). (Corresponding author: Lan Du.)

The authors are with the National Laboratory of Radar Signal Processing and the Collaborative Innovation Center of Information Sensing and Understanding, Xidian University, Xi'an 710071, China (e-mail: luli710071@163.com; dulan@mail.xidian.edu.cn; zcwang199009@163.com).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JSTARS.2018.2874128

Gaussian pyramids. By means of Kullback–Leibler divergence, Klein and Frintrap [10] regarded the multiscale contrast between the feature distributions of center and its surroundings as the center-surround difference.

Taking the global rarity as saliency cue, Duan *et al.* [11] measured the global rarity of each region via the spatially weighted dissimilarity. Wang *et al.* [12] computed the global spatially dissimilarity of superpixels to generate the superpixel-level saliency map, and then, fused it with another saliency map in the Bayesian framework. Hou *et al.* [13] obtained the spectral residual via analyzing the log spectrum of an image, and then, transformed the spectral residual to the spatial domain to generate the saliency map.

Taking the difference between targets and clutter as saliency cue, researchers extracted clutter templates at first, and then, computed the difference between every sample and the clutter templates as the value of saliency to highlight the targets [14]–[20]. Since human beings always tend to place targets of interest in the center of optical images, which is a validated reasonable and effective basic rule in optical images [21], Wei *et al.* [22] proposed that the boundary of optical images can be regarded as clutter, which is called the boundary prior. Therefore, patches or superpixels are usually extracted from the boundary of optical images as the clutter templates for the saliency detection in optical images. And the more different the sample to be detected and clutter templates are, the more likely the sample is target sample. As a method that regards the difference between targets and clutter as saliency cue, the sparse-reconstruction-based saliency detection method [18], [19] in optical images is flourishing with the development of sparse reconstruction. On the basis of the boundary prior, clutter templates are extracted from the boundary of optical images, and then, construct a clutter dictionary. Based on the assumption of sparse reconstruction that a sample can be well reconstructed by a small number of templates from the same class, the reconstruction residual of the sample is regarded as the measurement of saliency that reflects the difference between the sample and the clutter templates.

The sparse-reconstruction-based saliency detection methods have demonstrated great performance in optical images [18], [19]. Therefore, we expect to introduce the sparse-reconstruction-based saliency detection to target detection in SAR images. On the one hand, the clutter information from clutter templates can be fully utilized. On the other hand, the clutter statistical modeling can be avoided. However, the sparse-reconstruction-based saliency method cannot be directly applied to the SAR images. Two important reasons are as follows. 1) The boundary prior which is used to extract clutter templates in optical images is not suitable for SAR images. The optical images are usually small-sized simple images and humans tend to place objects of interest in the center of optical images. However, the target detection in SAR images generally aims to large complex scenes. Moreover, it is not possible to intentionally place the targets in the center of the SAR images. 2) Different from optical images, the intensities of pixels are fluctuant due to the influence of speckle noise. Only utilizing the intensity information at the pixel-level is insufficient.

According to the aforementioned analyses, in this paper, we propose a novel target detection method based on dual-domain

sparse reconstruction saliency in SAR images. More specifically, the main contributions of the proposed target detection method are as follows.

- 1) *We introduce the sparse-reconstruction-based saliency to the target detection in SAR images.* The sparse-reconstruction-based saliency in the proposed target detection method not only makes full use of the clutter information from clutter templates, but also avoids the clutter statistical modeling, where it is difficult to select an appropriate clutter distribution model especially in complex scenes and an inappropriate clutter model will lead to poor detection performance.
- 2) *We utilize not only the intensity information in the image domain but also the structure information in feature domain, considering the speckle noise in SAR images.* In the image domain, the pixel-level intensity descriptor is utilized for generating the image-domain saliency map. In feature domain, the superpixel-level locality-constrained linear coding (LLC) descriptor based on pixel-level scale invariant feature transform (SIFT) features is utilized for generating the feature-domain saliency map, where the LLC can describe the spatial and structural information at superpixel level. Through integrating the saliency maps from the two domains, we take advantages of both the intensity information and the structure information to make the proposed target detection method more robust to speckle noise.
- 3) *We employ different dictionary constructions in the image domain and the feature domain, respectively, for sparse reconstruction, where the clutter templates, i.e., the elements of dictionaries, are obtained from some pure clutter SAR images.* In the image domain, we construct the dictionary only based on clutter templates, which are obtained through taking samples from the pure clutter SAR images. Since the potential target templates, obtained from the image-domain saliency map, can provide additional target information for reducing false alarms in saliency detection, we construct not only the clutter dictionary with clutter templates but also the mixed dictionary with both clutter templates and potential target templates for sparse reconstruction in the feature domain.

The aforementioned advantages can make the proposed method achieving better target detection performance. And the experiments on the miniSAR real data and the RADARSAT-2 data demonstrate that the proposed target detection method outperforms other compared methods in terms of detection performance.

The remainder of this paper is organized as follows. Section II introduces the proposed novel target detection method. Section III gives the experimental results and analysis based on the miniSAR real data and the RADARSAT-2 data. Finally, Section IV concludes this paper.

II. SAR TARGET DETECTION METHOD BASED ON THE DUAL-DOMAIN SPARSE RECONSTRUCTION SALIENCY

Fig. 1 shows the flowchart of the proposed target detection method in SAR images. As shown in Fig. 1, the framework of

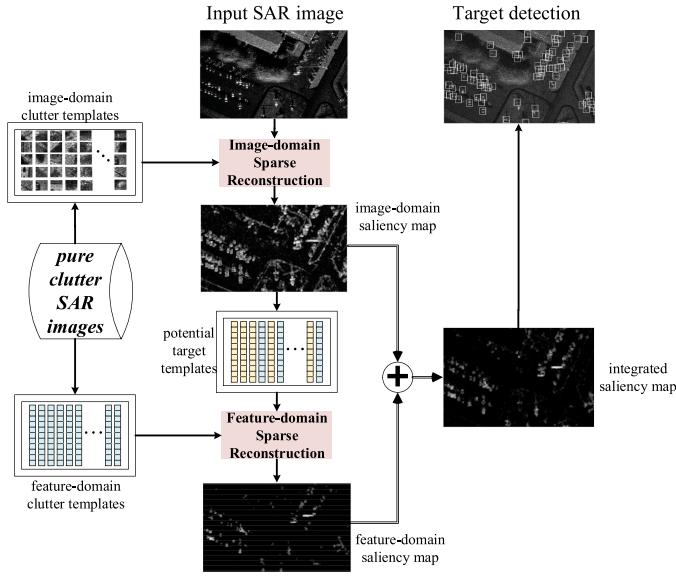


Fig. 1. Flowchart of the proposed target detection method in SAR images.

the proposed target detection method is briefly introduced as follows.

- 1) *Acquirement of clutter templates*: A great deal of clutter samples can be obtained through taking samples from the pure clutter SAR images, and then, some representative clutter samples are selected as clutter templates for subsequent works.
- 2) *Construction of image-domain saliency map*: The image-domain saliency map is constructed via the sparse reconstruction on clutter templates in image domain with the pixel-level intensity information as the descriptor.
- 3) *Construction of feature-domain saliency map*: The feature-domain saliency map is constructed via the sparse reconstruction on both clutter and potential target templates in the feature domain with the superpixel-level LLC of pixel-level SIFT features as the descriptor, where the potential target templates are obtained on the basis of the image-domain saliency map.
- 4) *Fusion of saliency maps and location of targets*: Since both of the intensity information in image domain and the structural information in the feature domain are significant for target detection, it is better to take the intensity information and the structural information into consideration simultaneously. Therefore, the image-domain saliency map and the feature-domain saliency map are integrated to generate the integrated saliency map. Then, the winner-take-all (WTA) [7], [23] approach is employed to detect targets from the integrated saliency map.

Among the aforementioned four steps, 1), 2), and 3) are crucial for the proposed target detection method, and the details of the first three steps are introduced concretely in the following sections.

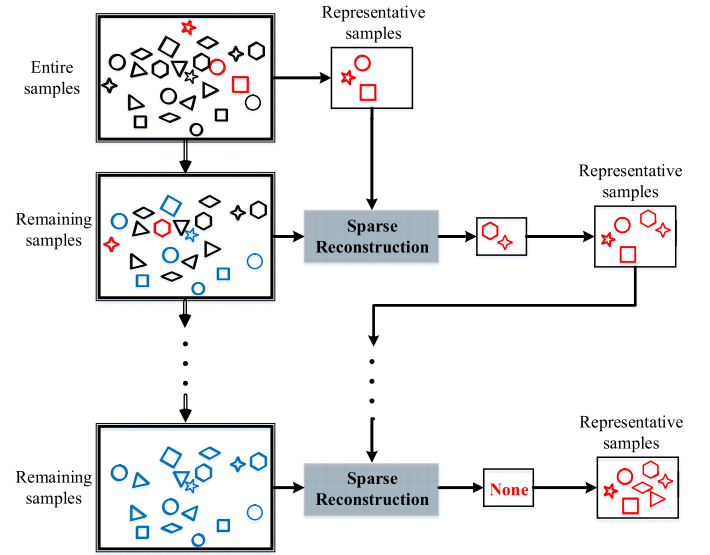


Fig. 2. Flowchart of the sample selection method. In the flowchart, the red ones stand for the selected representative samples, the blue ones stand for the samples that can be well sparse reconstructed, and the black ones stand for the samples that cannot be well sparse reconstructed.

A. Acquirement of Clutter Templates

In optical images, there is an effective basic rule that clutter templates can be extracted from the boundary of optical images, called the boundary prior. However, the boundary prior is obviously not suitable for SAR images. Therefore, we draw support from the pure clutter SAR images to generate clutter templates for the target detection in SAR images rather than the boundary prior.

We take samples from the pure clutter SAR images to generate a great deal of clutter samples. Nevertheless, the volume of clutter samples is enormous and the clutter samples have some redundancies.

Therefore, we apply a sample selection method to select representative clutter samples from the entire clutter samples [24]. We utilize the sparse reconstruction residual of a sample to measure whether the representative sample subset contains the same kind samples as this sample. When the residual of a sample is large, the representative sample subset does not contain the same kind samples with this sample, and then, this sample will be regarded as the representative sample. The procedure of the sample selection method is shown in Algorithm 1 and the visualization is shown in Fig. 2.

Through iterative selection, the representative clutter samples can be selected from the entire clutter samples. And then, the representative clutter samples can be regarded as the clutter templates for the subsequent sparse-reconstruction-based saliency detection.

B. Construction of the Image-Domain Saliency Map

In the image domain, we take the pixel-level intensity information as descriptor and utilize the sparse reconstruction theory to generate the image-domain saliency map. The pixel-level intensity information in original image, called image-domain

Algorithm 1. The Procedure of the Sample Selection Method.

Input: ψ -The set of entire samples.

Output: φ -The subset of representative samples.

Initialization:

Set φ as an empty set.

Set the number of randomly selected samples n_{rs} .

Set the number of selected samples in iterations n_{is} .

1. Put n_{rs} samples randomly selected from ψ into φ .

Repeat:

2. Update $\psi_{rem} = \psi - \varphi$.

3. Regard φ as the dictionary to sparsely reconstruct the samples in ψ_{rem} .

4. Sort the reconstruction residual of samples in ψ_{rem} from big to small and select n_{is} samples with the largest residual.

5. Put the n_{is} samples into φ .

Until: the convergence conditions are satisfied.

feature, is important for detection due to the pixel-level intensity differences between targets and clutter. As usually used in SAR images, the image-domain descriptor of a pixel $\mathbf{x} \in \mathbb{R}^2$ can be represented by a vector of intensity information in region $\Omega(\mathbf{x}) \in \mathbb{R}^{m \times m}$ centered by the pixel $\mathbf{x}$, where m is the length and width of region $\Omega(\mathbf{x})$. The procedure of the proposed saliency detection method in image domain is as follows.

1) With the image-domain descriptor, image-domain clutter templates are extracted from pure clutter SAR images to construct the clutter dictionary.

First, we obtain a great deal of image-domain clutter samples through a sliding window on the pure clutter SAR images. Then, the sample selection method, illuminated in Section II-A, is applied to select the image-domain representative clutter samples. Finally, the representative clutter samples construct the image-domain clutter dictionary $\mathbf{D}^I = [\mathbf{d}_1^I, \mathbf{d}_2^I, \mathbf{d}_3^I, \dots, \mathbf{d}_{N_I-1}^I, \mathbf{d}_{N_I}^I] \in \mathbb{R}^{m^2 \times N_I}$, where $\mathbf{d}_j^I \in \mathbb{R}^{m^2 \times 1}$, $j \in [1, N_I]$ is the image-domain clutter template, and N_I is the number of clutter template in the image-domain clutter dictionary.

2) Based on the image-domain clutter dictionary $\mathbf{D}^I$, we utilize sparse reconstruction to generate the image-domain saliency map.

The sparse-reconstruction-based saliency method takes the difference between targets and clutter as the saliency cue to highlight salient targets, which can make the clutter information in clutter templates fully utilized for better target detection and avoid the clutter statistical modeling. As depicted in Fig. 3, for each pixel $\mathbf{x}_i$ in SAR images to be detected, we represent its image-domain descriptor as $\mathbf{f}_I(\mathbf{x}_i)$, and then, encode the $\mathbf{f}_I(\mathbf{x}_i)$ based on the image-domain clutter dictionary $\mathbf{D}^I$. The sparse coding is formulated as

$$\alpha_i^* = \arg \min_{\alpha_i} \|\mathbf{f}_I(\mathbf{x}_i) - \mathbf{D}^I \alpha_i\|_2^2 + \lambda \|\alpha_i\|_1 \quad (1)$$

where $\|\cdot\|_1$ denotes the l_1 -norm, $\|\cdot\|_2$ denotes the l_2 -norm, and λ is a regularization parameter.

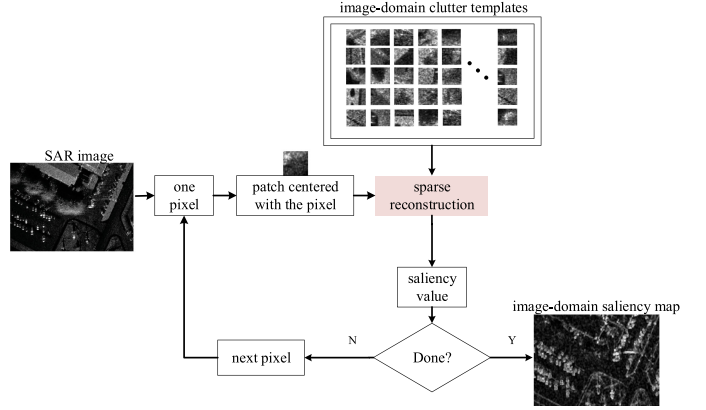


Fig. 3. Flowchart of the generation of the image-domain pixel-level saliency map.

Then, the sparse reconstruction residual is represented as

$$\varepsilon_i^I = \|\mathbf{f}_I(\mathbf{x}_i) - \mathbf{D}^I \alpha_i^*\|_2^2. \quad (2)$$

The sparse-reconstruction-based saliency method is based on the assumption that the samples can be reconstructed by some templates from the same kind. Therefore, based on the clutter templates, only the clutter samples in SAR images can be reconstructed well while the target samples in SAR images are reconstructed poorly. That is to say, the reconstruction residual ε_i^I , which reflects how well the reconstruction is performing can be regarded as the measurement of the salient value for the pixel $\mathbf{x}_i$. The smaller the reconstruction residual is, the less the saliency of the sample to be detected is, and vice versa. Therefore, we can obtain the image-domain saliency map Sal^I when the sparse reconstruction residual of every pixel in SAR images is computed.

C. Construction of the Feature-Domain Saliency Map

Due to the speckle noise, the intensities of pixels in the SAR images are fluctuant, making it difficult to obtain the clear borders of salient targets. Superpixels, which are small regions consisting of a series of spatially connected similar pixels through segmentation, can keep the borders of objects excellently and improve the computational efficiency [25]. Meanwhile, the dense SIFT feature that describes the gradient information can be a complement to the image-domain intensity information [26]. Therefore, in feature domain, we adopt the superpixels as the elementary units and extract the structural information from superpixels to make saliency detection more robust to speckle noise. To extract structural information from superpixel, we extract the dense SIFT in a fixed size region centered on superpixel [24], and then, encode the dense SIFT feature to midlevel feature through the LLC model [27] since it has been demonstrated that midlevel features can well catch the spatial and structural information.

Furthermore, the conventional sparse-reconstruction-based saliency detection only utilizes the clutter information from the clutter templates. However, the target information also plays an important role in highlighting targets. Many researchers extract potential target information from a given saliency map

to help subsequent saliency detection [16], [20]. Inspired the important role of potential target information for the saliency detection, we propose a sparse-reconstruction-based saliency detection method considering both clutter information and target information.

The procedure of the proposed saliency detection method in feature domain is as follows.

- 1) With the superpixel-level LLC feature as descriptor, we extract feature-domain clutter templates from the superpixels in the pure clutter SAR images.

First, the pure clutter SAR images are segmented into superpixels through the simple linear iterative clustering (SLIC) [28], one of the best segmentation algorithm for its boundary adherence and efficiency. Then, the LLC features are extracted from all of the superpixels as descriptors of feature-domain clutter samples. Finally, the sample selection method, illuminated in Section II-A, is applied to select some representative feature-domain clutter samples as the feature-domain clutter templates $[\mathbf{d}_1^{\text{FC}}, \mathbf{d}_2^{\text{FC}}, \dots, \mathbf{d}_{N_C}^{\text{FC}}] \in \mathbb{R}^{l \times N_C}$, where $\mathbf{d}_j^{\text{FC}} \in \mathbb{R}^{l \times 1}, j \in [1, N_C]$ is the feature-domain clutter template, N_C is the number of feature-domain clutter templates, and l is the dimension of the LLC feature.

- 2) With the superpixel-level LLC feature as descriptor, we extract potential target templates from the superpixels, which are salient in the image-domain saliency map.

First, the SAR image is segmented into superpixels through the SLIC algorithm [28], and then, select some potential target superpixels that are salient in the image-domain saliency map. Finally, we extract the LLC features from the potential target superpixels as the potential target templates $[\mathbf{d}_1^{\text{FT}}, \mathbf{d}_2^{\text{FT}}, \mathbf{d}_3^{\text{FT}}, \dots, \mathbf{d}_{N_S-1}^{\text{FT}}, \mathbf{d}_{N_S}^{\text{FT}}] \in \mathbb{R}^{l \times N_S}$, where $\mathbf{d}_j^{\text{FT}} \in \mathbb{R}^{l \times 1}, j \in [1, N_S]$ is the feature-domain potential target template, and N_S is the number of potential target templates.

- 3) Based on the clutter templates and the potential target templates, we construct two different dictionaries.

One of the dictionaries is constructed only by the clutter templates, which is represented as $\mathbf{D}^{\text{FC}} = [\mathbf{d}_1^{\text{FC}}, \mathbf{d}_2^{\text{FC}}, \mathbf{d}_3^{\text{FC}}, \dots, \mathbf{d}_{N_C-1}^{\text{FC}}, \mathbf{d}_{N_C}^{\text{FC}}] \in \mathbb{R}^{l \times N_C}$. The other dictionary contains both the clutter templates and the potential target templates, which is represented as $\mathbf{D}^{\text{FE}} = [\mathbf{d}_1^{\text{FC}}, \mathbf{d}_2^{\text{FC}}, \dots, \mathbf{d}_{N_C}^{\text{FC}}, \mathbf{d}_1^{\text{FT}}, \mathbf{d}_2^{\text{FT}}, \dots, \mathbf{d}_{N_S}^{\text{FT}}] \in \mathbb{R}^{l \times N_E}$, where $N_E = N_C + N_S$ is the number of templates in dictionary $\mathbf{D}^{\text{FE}}$.

- 4) The feature-domain saliency map is constructed via the sparse reconstruction on two different dictionaries.

For every superpixel $\mathbf{s}_i, i \in [1, N_S]$, we represent its LLC feature as $\mathbf{f}_F(\mathbf{s}_i)$ (similar to $\mathbf{d}_i^{\text{FT}}$). Our sparse-representation-based saliency method in feature domain, which uses two different dictionaries, also relies on this assumption that the sample to be detected can be linearly reconstructed well by a small number of templates from the same class. As depicted in Fig. 4, we encode the $\mathbf{f}_F(\mathbf{s}_i)$ based on two different dictionaries, $\mathbf{D}^{\text{FC}}$ and

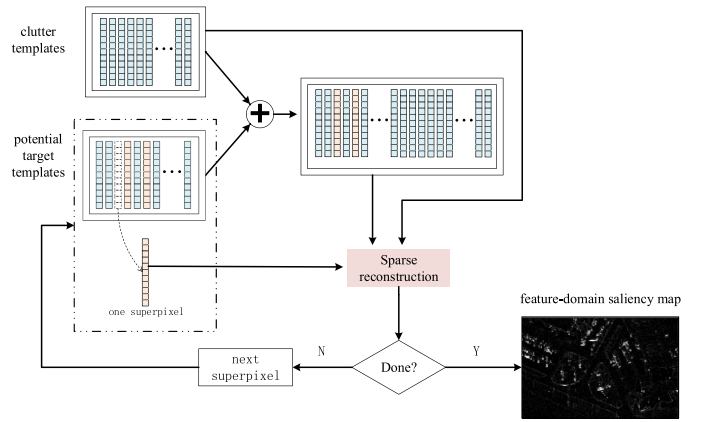


Fig. 4. Flowchart of the generation of the feature-domain superpixel-level saliency map.

$\mathbf{D}^{\text{FE}'}$. The sparse representations and reconstruction residuals under two dictionaries are, respectively, formulated as

$$\beta_i^* = \arg \min_{\beta_i} \|\mathbf{f}_F(\mathbf{s}_i) - \mathbf{D}^{\text{FC}} \beta_i\|_2^2 + \lambda \|\beta_i\|_1 \quad (3)$$

$$\varepsilon_i^{\text{FC}} = \|\mathbf{f}_F(\mathbf{s}_i) - \mathbf{D}^{\text{FC}} \beta_i^*\|_2^2 \quad (4)$$

$$\gamma_i^* = \arg \min_{\gamma_i} \|\mathbf{f}_F(\mathbf{s}_i) - \mathbf{D}^{\text{FE}'} \gamma_i\|_2^2 + \lambda \|\gamma_i\|_1 \quad (5)$$

$$\varepsilon_i^{\text{FE}} = \|\mathbf{f}_F(\mathbf{s}_i) - \mathbf{D}^{\text{FE}'} \gamma_i^*\|_2^2 \quad (6)$$

where $\mathbf{D}^{\text{FE}'}$ is the same with $\mathbf{D}^{\text{FE}}$ except that $\mathbf{D}^{\text{FE}'}$ does not contain the $\mathbf{d}_i^{\text{FT}}$.

Since the clutter templates consist in the $\mathbf{D}^{\text{FC}}$ and $\mathbf{D}^{\text{FE}'}$ simultaneously, a clutter sample can be sparsely reconstructed by each of the two dictionaries. That is to say, the difference between the two reconstruction residuals under the two dictionaries will be small if the sample to be detected is a clutter sample. On the other hand, if the sample to be detected is a target sample, it can be sparsely reconstructed by the dictionary $\mathbf{D}^{\text{FE}'}$ rather than the dictionary $\mathbf{D}^{\text{FC}}$, since the target templates only consist in $\mathbf{D}^{\text{FE}'}$. That is to say, the difference between the reconstruction residual under dictionary $\mathbf{D}^{\text{FE}'}$ and that under dictionary $\mathbf{D}^{\text{FC}}$ will be large when the sample to be detected is a target sample. Therefore, the differences between the sparse reconstruction residuals under two dictionaries can be the measurement of saliency. The greater the difference between the two representation residuals is, the more likely the sample to be detected is a target sample.

The difference between the two sparse representation residuals under two different dictionaries is represented as

$$\Delta \varepsilon_i^F = |\varepsilon_i^{\text{FC}} - \varepsilon_i^{\text{FE}}|. \quad (7)$$

The $\Delta \varepsilon_i^F$ represent the salient value of the superpixels $\mathbf{s}_i, i \in [1, N_S]$ and the salient value of pixel that contains in the superpixel $\mathbf{s}_i$ is set to be equal to that of the superpixel. Therefore, we can get the feature-domain saliency map Sal^F .

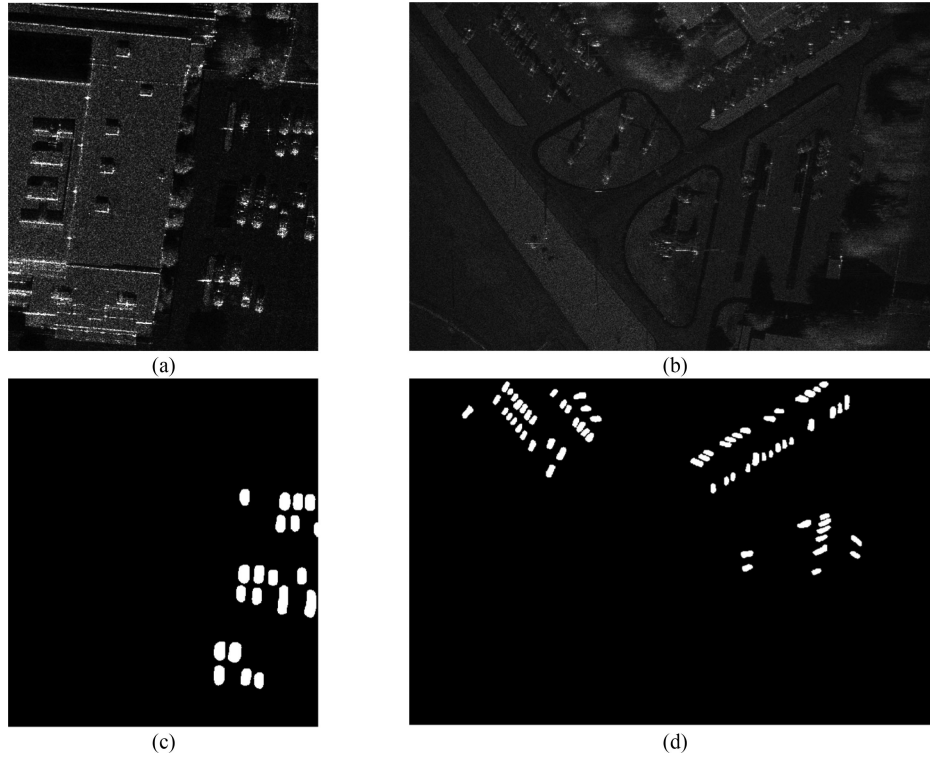


Fig. 5. Original SAR images and their corresponding pixel-level ground truths. (a) and (c) Original miniSAR FARAD image and its corresponding pixel-level ground truths. (b) and (d) Original miniSAR image and its corresponding pixel-level ground truths.

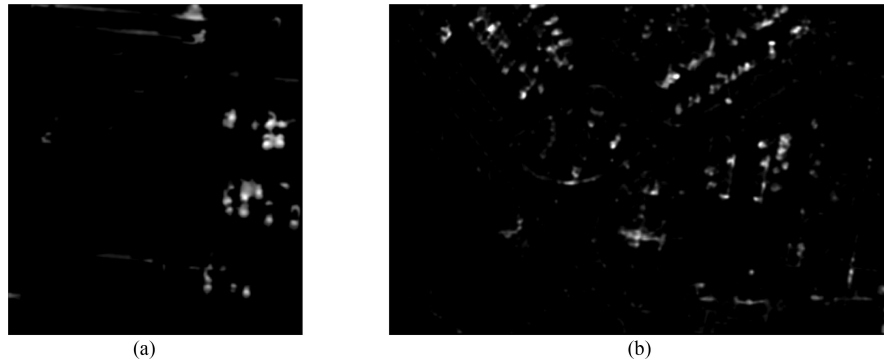


Fig. 6. Saliency maps obtained from the proposed method. (a) Saliency map on the miniSAR FARAD image. (b) Saliency map on the miniSAR image.

III. EXPERIMENTAL RESULTS AND ANALYSIS

The measured miniSAR FARAD dataset and miniSAR dataset are acquired by U.S. Sandia National Laboratories in 2015 and 2005, respectively. We select one miniSAR FARAD image, with the resolution of $0.1 \text{ m} \times 0.1 \text{ m}$ and the size of 960×830 pixels, and one miniSAR image, with the resolution of $0.1 \text{ m} \times 0.1 \text{ m}$ and the size of 1638×2510 pixels, for our experiments. Fig. 5 shows the original miniSAR FARAD image, the original miniSAR image and their corresponding pixel-level ground truths, respectively, where the vehicles are the targets of interest and the others are regarded as clutter.

In the proposed method, we extract different descriptors in the image domain and the feature domain. For the image-domain descriptor, the length and width m of region $\Omega(\mathbf{x}) \in \mathbb{R}^{m \times m}$ is set to 9 pixels. For the feature-domain descriptor, the size of the fixed size region for extracting the dense SIFT feature is

set to 64×64 pixels, and the dimension of the LLC feature l is set to 1 024 empirically. Based on descriptors, we employ sparse reconstruction to generate saliency maps. In the procedure of sparse reconstruction, there is a regularization parameter λ , which is empirically set to 0.01 [17]–[19]. Furthermore, two situations are regarded as convergence conditions in the procedure of representative sample selection. First, the number of selected representative samples is more than 2000. Second, the average reconstruction residual of sparse representation is lower than 0.5%. We iteratively run to select samples until any of the two convergence conditions are reached.

A. Saliency Map Results

As shown in Fig. 6, the vehicle targets are much brighter than most clutter, e.g., trees and buildings, in saliency maps. Therefore, we can easily extract the vehicle targets from the

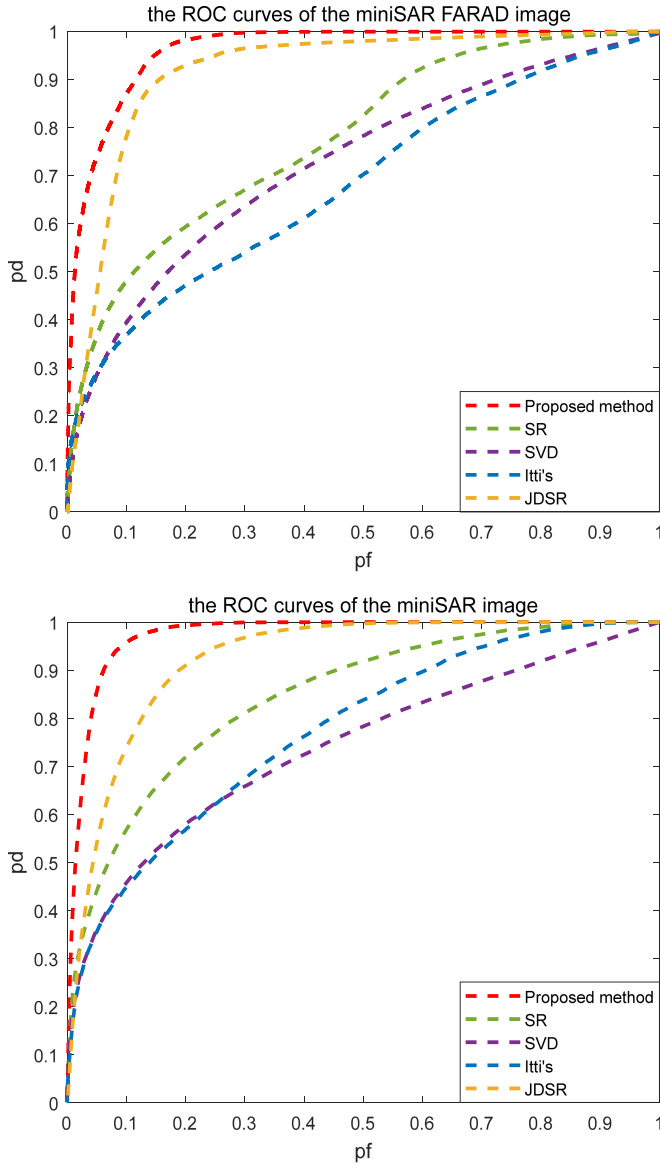


Fig. 7. ROC curves of different saliency detection methods on the miniSAR FARAD image and the miniSAR image. Here, the red line represents the ROC curve of the proposed method.

saliency maps without a large number of false alarms. In the following, we will quantitatively compare the proposed method with other methods.

B. Evaluation of Saliency Maps

In order to evaluate the effectiveness of saliency models, we employ the receiver operating characteristic (ROC) curve [29] and the area under ROC curve (AUC) [30]. Given a saliency map, by plotting the true positive rate against the false positive rate at various threshold settings, the ROC curve is achieved, and then, its underneath area AUC can be calculated.

In Fig. 7, we compare the proposed method with other saliency methods in terms of ROC curves and Table I gives their corresponding values of the AUC. The SR [13] is the

TABLE I
AUCs OF DIFFERENT SALIENCY DETECTION METHODS

Image	Our Method	SR	SVD	Itti's	JDSR
The miniSAR FARAD image	0.9624	0.7786	0.7272	0.6835	0.9176
The miniSAR image	0.9738	0.8450	0.7406	0.7732	0.9244

method of spectral residual. The SVD [8] is a visual attention method based on SVD. The Itti's [7] is the saliency-based visual attention model proposed by Itti *et al.* The JDSR, modified from [26], employs jointly dynamic sparse representation, by which the intensity information and the gradient information are combined, to generate a saliency map. Different from [26], which is a target recognition method and needs training samples of each target category, here the JDSR, as a saliency method, only utilizes the clutter samples.

Fig. 7 shows the ROC curves of different saliency detection methods. The closer to the upper left the ROC curve is, the better the performance of the saliency detection performance is. In Fig. 7, the red line represents the ROC curve of the proposed method, and we can clearly see that the proposed method outperforms all the other compared saliency methods on both the miniSAR FARAD image and the miniSAR image. Table I gives the values of the AUC of different saliency detection methods. The larger the AUC is, the better the performance of the saliency detection is. We can observe from Table I that the value of the AUC of the proposed method are 0.9624 on the miniSAR FARAD image and 0.9738 on the miniSAR image, respectively. Compared with the Itti's method, the SR method, the SVD method, and the JDSR method, the proposed method works best.

C. Evaluation of Target Detection Results

In this section, we compare the proposed method with the conventional CFAR methods that are usually used for target detection in SAR images. The conventional CFARs for comparison contain the Gaussian-CFAR (i.e., two parameters CFAR) [1] that assumes the clutter obeying Gaussian statistical distribution and the Gamma-CFAR [2] that assumes the clutter obeying the Gamma statistical distribution. However, it is difficult to select an appropriate distribution especially in complex scenes and an inappropriate clutter model will lead to poor detection performance. The proposed method avoids the clutter statistical modeling while makes full use of clutter information.

Fig. 8 shows the intuitional target detection results of the proposed method, the Gaussian-CFAR and the Gamma-CFAR on the miniSAR FARAD image and the miniSAR image. As can be seen from Fig. 8, the proposed target detection method has fewest false alarms and missing alarms. For quantitative analysis, we evaluate the target detection results in terms of recall, precision, F1-score, and the figure of merit (FoM). The

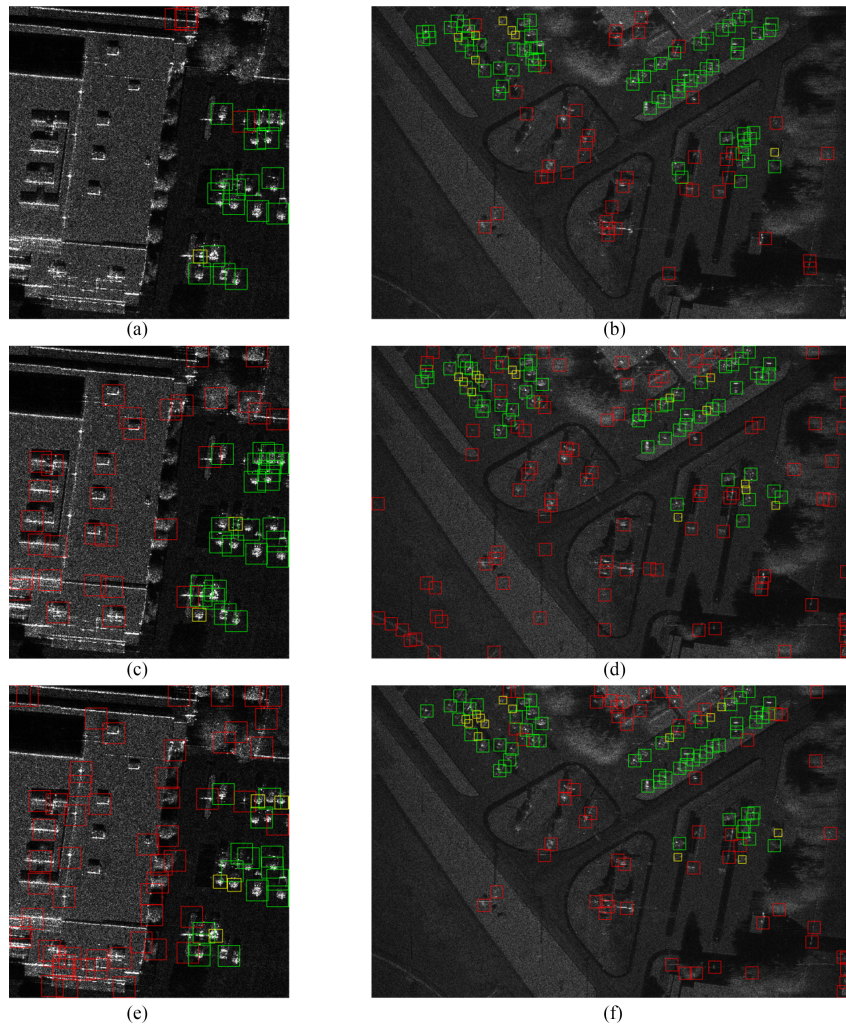


Fig. 8. Target detection results on the miniSAR FARAD image and the miniSAR image, where green rectangles represent the detected target chips, red rectangles represent false alarms, and yellow rectangles represent the missing alarms. (a) and (b) Target detection results of the proposed method, respectively. (c) and (d) Target detection results of the Gaussian-CFAR, respectively. (e) and (f) Target detection results of the Gamma-CFAR, respectively.

TABLE II
OVERALL EVALUATION OF DIFFERENT TARGET DETECTION METHODS

	Target amounts	Missing alarms	False alarms	Chip amounts	Recall	Precision	F1-score	FoM
Our Method	85	8	44	128	0.9059	0.6563	0.7611	0.5969
Gaussian CFAR		16	126	203	0.8235	0.3793	0.5194	0.3270
Gamma CFAR		19	117	189	0.7765	0.3810	0.5112	0.3267

definitions of those evaluation criteria are as follows:

$$\text{recall} = \frac{\text{the number of detected targets}}{\text{the total number of targets}} \quad (8)$$

$$\text{precision} = \frac{\text{the number of target chips}}{\text{the total number of chips}} \quad (9)$$

$$F1 - \text{score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (10)$$

$$\text{FoM} = \frac{\text{the number of detected targets}}{\text{the number of targets} + \text{the number of false alarms}}. \quad (11)$$

The higher the value of recall, precision, F1-score, and FoM are, the better the performance of target detection is, and vice versa.

Table II is the overall evaluation of the proposed method, the Gaussian CFAR, and the Gamma CFAR. It is clearly seen from

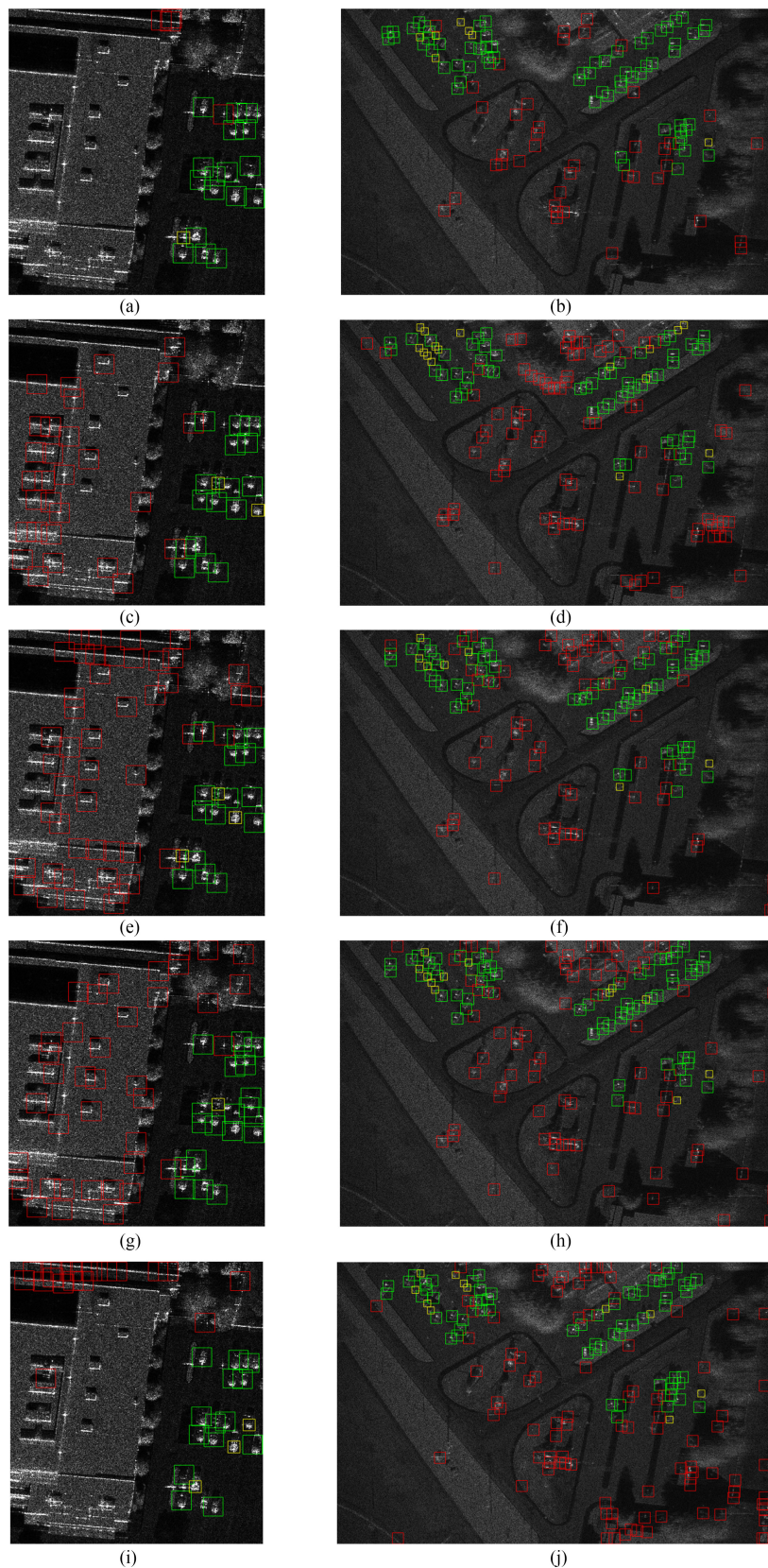


Fig. 9. Target detection results of the proposed saliency detection method and other saliency detection methods on the miniSAR FARAD image and the miniSAR image, where green rectangles represent the detected target chips, red rectangles represent false alarms and yellow rectangles represent missing alarms. (a) and (b) Target detection results of the proposed method, respectively. (c) and (d) Target detection results of the Itti's method, respectively. (e) and (f) Target detection results of the SVD method, respectively. (g) and (h) Target detection results of the SR method, respectively. (i) and (j) Target detection results of the JDSR method, respectively.

TABLE III
OVERALL EVALUATION OF DIFFERENT SALIENCY METHODS FOR TARGET DETECTION

	Target amounts	Missing alarms	False alarms	Chip amounts	Recall	Precision	F1-score	FoM
Our Method	85	8	44	128	0.9059	0.6563	0.7611	0.5969
Itti's Method		18	110	181	0.7882	0.3923	0.5239	0.3436
SVD Method		11	113	193	0.8706	0.4145	0.5616	0.3737
SR Method		13	115	197	0.8470	0.4162	0.5581	0.3600
JDSR Method		14	114	192	0.8353	0.4063	0.5467	0.3568

TABLE IV
TIME COSTS OF DIFFERENT TARGET DETECTION METHODS ON THE miniSAR IMAGE

Methods	Our Method	Itti's method	SR method	SVD method	JDSR method	Gaussian CFAR	Gamma CFAR
Average Time Costs (second)	5732.8	0.7	1.6	4.7	8297.5	32.3	1278.1

Table II that the proposed method has highest recall, precision, F1-score, and FoM, that is to say, the proposed method has the best target detection performance compared with the Gaussian CFAR and the Gamma CFAR.

D. Comparisons With Other Saliency Detection Methods for Target Detection

Through the application of the WTA [7], [23] approach, the results of target detection can be obtained from a saliency map. In this section, we compare the results of the target detection of the proposed method and other saliency detection methods intuitively and quantitatively. The saliency detection methods for comparison contain the Itti's method [7], the SVD method [8], the SR method [13], and the JDSR method [26].

Fig. 9 shows the target detection results of the proposed method and other saliency detection methods on the miniSAR FARAD image, and the miniSAR image. Compared with the other saliency methods, the proposed method achieves fewest false alarms and missing alarms. For quantitative analysis, we compare the proposed method with other saliency detection methods in terms of recall, precision, F1-score, and FoM in Table III. As shown in Table III, the recall and the precision of the proposed method are the highest among all the methods. From F1-score and FoM, we can see that the proposed method achieves the best detection performance.

E. Analysis on the Time Costs and the Complexities

1) *Time Costs*: In Table IV, the time costs of the proposed method, the Itti's method, the SR method, the SVD method, the JDSR method, the Gaussian-CFAR, and the Gamma-CFAR on the miniSAR image are given. All computations are performed on Intel Core CPU i7-4900MQ at 2.8 GHz. From Table IV, we can see that the time cost of the proposed method is larger

TABLE V
COMPLEXITY OF THE PROPOSED METHOD AND THE ITTI'S METHOD

	Our Method	Itti's method
Time Complexity	$O(\max(Nl_1^2 k_1^2, N, Ml_2^2 k_2^2))$	$O(\max(N, T \cdot \frac{N}{2^{2/\min}}))$
Asymptotic Time Complexity	$O(N)$	$O(N)$
Space Complexity	$O(2N + M + 5 \max(L_1, L_2))$	$O(8N)$

than the Itti's method, the SR method, the SVD method, the Gaussian-CFAR, and the Gamma-CFAR and lower than the JDSR method.

The reasons for the proposed method having a relative high time cost can be summarized as follows. First, the proposed method extract the LLC descriptor based on SIFT features in the feature domain. The feature extraction of the LLC feature is complex rather than simple filtering processes (e.g., Gaussian filtering and Gabor filtering) in the Itti' method and the SVD method, and Fourier Transform in the SR method. Second, in the image domain, the proposed method needs to sparse reconstruct each pixel in an image. The computational complexity of sparse reconstruction is higher than that of computing the parameters of the statistical model in the Gaussian-CFAR and the Gamma-CFAR. Third, besides the sparse reconstruction for each pixel in image domain, the proposed method also sparse reconstructs each superpixel in the feature domain.

However, comparing with the JDSR method, the proposed method has a lower time cost. The JDSR method and the proposed method both require sparse reconstruction in the image domain and the feature domain. In image domain, the JDSR method and the proposed method sparse reconstruct each pixel

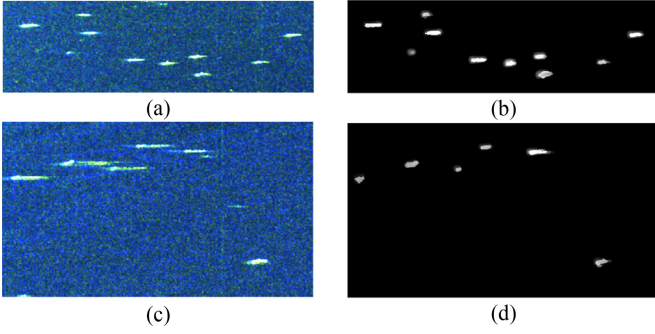


Fig. 10. Original RADARSAT-2 POLSAR subimages and their corresponding saliency maps that are obtained from the proposed method on Images I and II. (a) and (b) Original Image I and its corresponding saliency map, respectively. (c) and (d) Original Image II and its corresponding saliency map, respectively. Ships are the targets of interest and they are highlighted in the saliency maps.

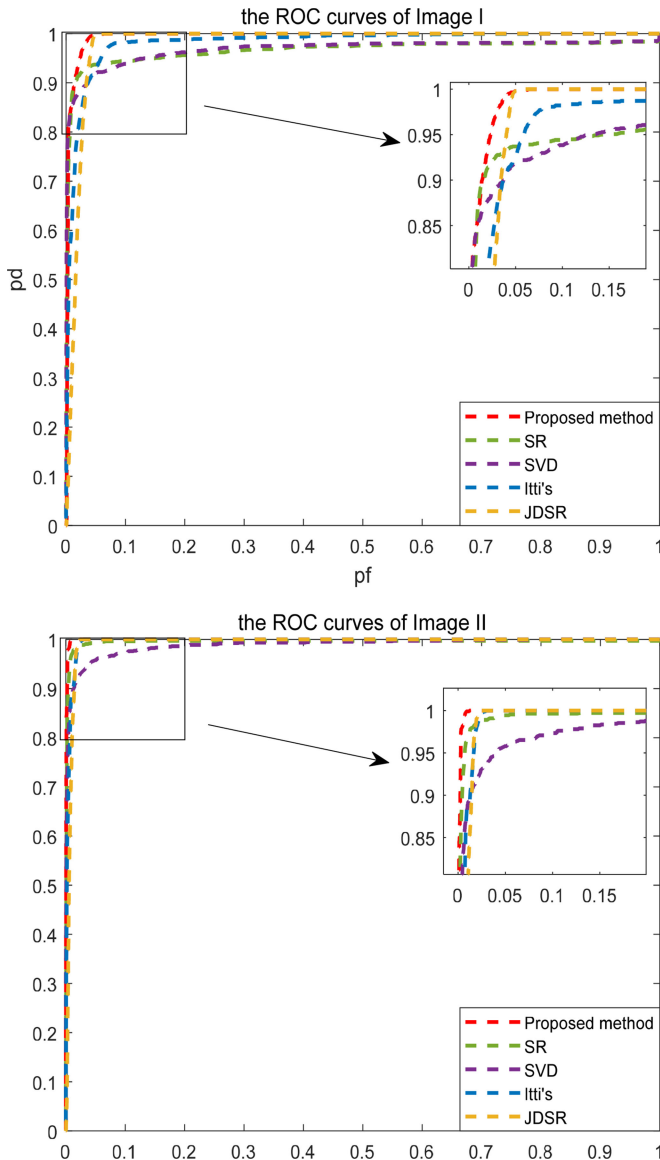


Fig. 11. ROC curves of different saliency detection methods on Images I and II. Here, the red line represents the ROC curve of the proposed method.

TABLE VI
AUCs OF DIFFERENT SALIENCY DETECTION METHODS ON IMAGES I AND II

Image	Our Method	SR	SVD	Itti's	JDSR
Image I	0.9948	0.9665	0.9682	0.9826	0.9824
Image II	0.9992	0.9953	0.9888	0.9960	0.9931

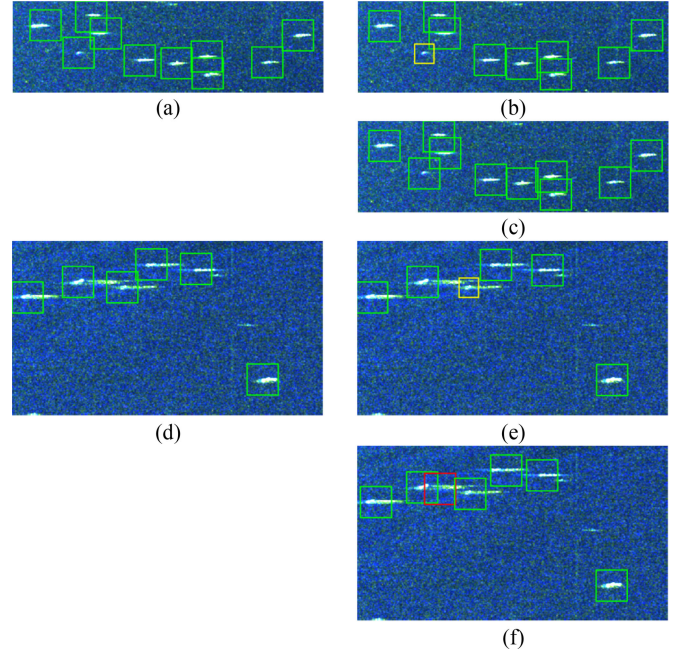


Fig. 12. Target detection results on Images I and II, where green rectangles represent the detected target chips, red rectangles represent false alarms, and yellow rectangles represent the missing alarms. (a) and (d) Target detection results of the proposed method, respectively. (b) and (e) Target detection results of the Gaussian-CFAR method, respectively. (c) and (f) Target detection results of the Gamma-CFAR method, respectively.

in an image. In feature domain, the JDSR method still regards each pixel as an elementary unit to be sparse reconstructed; while the proposed method regards each superpixel segmented through the SLIC as an elementary unit to be sparse reconstructed. The amount of superpixels in an image is much less than the amount of pixels. Therefore, the time cost of the proposed method is lower than the JDSR method.

Furthermore, in the proposed method, the sparse reconstruction of one unit, i.e., pixel in image domain and superpixel in feature domain, is not affected by other units, that is to say, the sparse reconstruction of every unit's is independent. Therefore, it can be implemented by using a parallel algorithm to reduce the time consumption.

2) *Complexities*: We further analyze the complexities of the proposed method compared with the classical Itti's method that has the minimum time cost among all the methods.

Table V gives the time complexities, the asymptotic time complexities, and the space complexities of the proposed method and the Itti's method, where N is the number of pixels in an image, M is the number of salient superpixels, l_1 and l_2 are

TABLE VII
OVERALL EVALUATION OF DIFFERENT TARGET DETECTION METHODS ON IMAGES I AND II

	Target amounts	Missing alarms	False alarms	Chip amounts	Recall	Precision	F1-score	FoM
Our Method	16	0	0	16	1	1	1	1
Gaussian CFAR		2	0	14	0.875	1	0.9333	0.8750
Gamma CFAR		0	1	17	1	0.9411	0.9696	0.9411

image-domain and feature-domain feature dimensions, respectively, k_1 and k_2 are the image-domain sparsity and feature-domain sparsity, respectively, L_1 and L_2 are the number of image-domain and feature-domain dictionary atoms, respectively, $I_{\min}$ is the minimum center scale, and T is the number of surrounding scale. From Table V, we can see that the time complexity of the proposed method is larger than that of the Itti's method, due to $l_1^2 k_1^2 > \frac{T}{2^{2I_{\min}}}$ in our experiments. Nevertheless, the asymptotic time complexity of the proposed method and that of the Itti's method are both $O(N)$, that is to say, this two methods are both linear in the number of pixels N . In the space complexity, the proposed method is lower than the Itti's method due to $L_1 \simeq L_2 \ll N$, $M \ll N$ in our experiments.

F. Extension of the Proposed Target Detection Method in RADARSAT-2 POLSAR Images

In this section, we extend to analyze the proposed target detection method in RADARSAT-2 POLSAR images. Here, we select two subimages from RADARSAT-2 data for our experiments, and refer to them as Images I and II, respectively.

Different from the miniSAR images, the POLSAR images are quad-polarimetric images. Motivated by [31] and [32], the dual-domain sparse reconstruction is applied to HH, HV, and VV, respectively, where H stands the horizontal polarimetric channel and V for the vertical polarimetric channel. Then, the saliency maps in HH, HV, and VV are fused to obtain the final saliency map to jointly exploit the polarimetric information. Finally, the WTA is applied to the final saliency map to detect targets. For the sake of fairness, other methods for comparison also employ similar scheme.

1) *Intuitional Saliency Map Results via the Proposed Method:* Fig. 10 shows the two original RADARSAT-2 POLSAR subimages and their corresponding saliency maps via the proposed method. From Fig. 10, we can see that the targets of interest, ships, have been highlighted in the saliency maps.

2) *Evaluation of the Saliency Map Results Compared With Other Saliency Methods:* In Fig. 11, we compare the proposed method with the Itti's method, the SR method, the SVD method, and the JDSR method in terms of ROC curves where the red line represents the ROC curve of the proposed method. For the sake of fairness, other methods for comparison also employ a similar scheme with the proposed method that three saliency maps are obtained, respectively, in HH, HV, and VV channels, and then, fused to obtain the final saliency map. And Table VI gives their corresponding values of the AUC. From Fig. 11, we can clearly

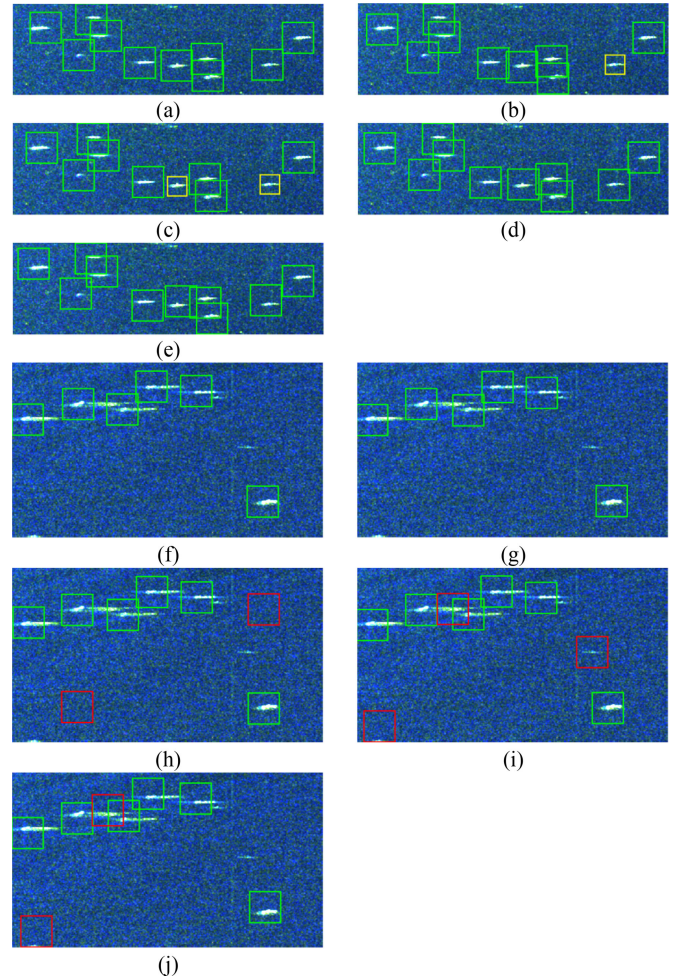


Fig. 13. Target detection results on Images I and II, where green rectangles represent the detected target chips, red rectangles represent false alarms, and yellow rectangles represent the missing alarms. (a) and (f) Target detection results of the proposed method, respectively. (b) and (g) Target detection results of the Itti's method, respectively. (c) and (h) Target detection results of the SVD method, respectively. (d) and (i) Target detection results of the SR method, respectively. (e) and (j) Target detection results of the JDSR method, respectively.

see that the proposed method outperforms all the other compared saliency methods. We can observe from Table VI that the value of the AUC of the proposed method are highest compared with other saliency methods.

3) *Evaluation of the Target Detection Results Compared With Other Target Detection Methods:* Fig. 12 shows the intuitional target detection results of the proposed method, the

TABLE VIII
OVERALL EVALUATION OF DIFFERENT SALIENCY METHODS FOR TARGET DETECTION ON IMAGES I AND II

	Target amounts	Missing alarms	False alarms	Chip amounts	Recall	Precision	F1-score	FoM
Our Method	16	0	0	16	1	1	1	1
Itti's Method		1	0	15	0.9357	1	0.9686	0.9375
SVD Method		2	2	16	0.875	0.875	0.875	0.7778
SR Method		0	3	19	1	0.8421	0.9142	0.8421
JDSR Method		0	2	18	1	0.8888	0.9411	0.8888

Gaussian-CFAR and the Gamma-CFAR on Images I and II, respectively. As can be seen from Fig. 12, the proposed target detection method has fewer false alarms and missing alarms compared with the Gaussian-CFAR and the Gamma-CFAR. And Table VII is the quantitative overall evaluation of the target detection results in terms of recall, precision, F1-score, and FoM on Images I and II. It is clearly seen from Table VII that the recall of the proposed method is the same with the Gamma-CFAR, higher than the Gaussian method. The precision of the proposed method is the same with the Gaussian-CFAR, higher than the Gamma-CFAR. From F1-score and FoM, comprehensive evaluation criteria, we can see that the proposed method achieves the best detection performance.

4) *Evaluation of the Target Detection Results Compared With Other Saliency Methods:* Fig. 13 shows the intuitional target detection results of the proposed method, the Itti's method, the SVD method, the SR method, and the JDSR method on Images I and II, respectively. As can be seen from Fig. 13, the proposed target detection method has fewer false alarms and missing alarms compared with the other saliency methods.

For quantitative analysis, we compare the proposed method with the Itti's method, the SVD method, the SR method, and the JDSR method in terms of recall, precision, F1-score, and FoM in Table VIII. The results in Table VIII are the overall results on Images I and II. As shown in Table VIII, the recall of the proposed method is the same with the SR method and the JDSR method, higher than the Itti's method and the SVD method. The precision of the proposed method is the same with the Itti's method, higher than the SVD method, the SR method, and the JDSR method. From F1-score and FoM, we can see that the proposed method achieves the best detection performance.

From the experiments and analysis of the proposed method for detecting targets on RADARSAT-2 POLSAR images, we can see that the proposed method has good performance not only in the miniSAR real data but also in the RADARSAT-2 POLSAR data.

IV. CONCLUSION

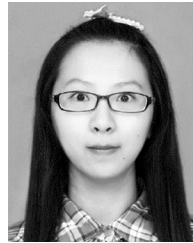
In this paper, we propose a novel target detection method based on the dual-domain sparse reconstruction saliency in SAR images. Through introducing the sparse-reconstruction-based saliency into the target detection in SAR images, the

proposed method avoids the clutter statistical modeling while makes full use of the clutter information. Furthermore, considering the speckle noise, the proposed method utilizes not only the intensity information in the image domain but also the structure information, i.e., LLC descriptor based on SIFT features, in the feature domain. The qualitative and quantitative experimental results on the miniSAR real data and the RADARSAT-2 POLSAR data have demonstrated that the proposed target detection method can achieve better detection performance than other compared methods.

REFERENCES

- [1] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 1, pp. 234–244, Jan. 1993.
- [2] C. Alberola-Lopez, J. Casar-Corredera, and G. De Miguel-Vela, "Object CFAR detection in gamma-distributed textured-background images," *Proc. IEEE*, vol. 146, no. 3, pp. 130–136, Jun. 1999.
- [3] C. Rother, V. Kolmogorov, and A. Blake, "GrabCut": Interactive foreground extraction using iterated graph cuts," *ACM Trans. Graph.*, vol. 23, pp. 309–314, 2004.
- [4] H. Wang, F. Xu, and S. Chen, "Saliency detector for SAR images based on pattern recurrence," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 7, pp. 2891–2900, Jul. 2016.
- [5] D. Zhao, Y. Ma, Z. Jiang, and Z. Shi, "Multiresolution airport detection via hierarchical reinforcement learning saliency model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 2855–2866, Jun. 2017.
- [6] L. Zhang and Y. Zhang, "Airport detection and aircraft recognition based on two-layer saliency model in high spatial resolution remote-sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1511–1524, Apr. 2017.
- [7] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 11, pp. 1254–1259, Nov. 1998.
- [8] S. Liu, Z. Cao, and J. Li, *A SVD-Based Visual Attention Detection Algorithm of SAR Image*. Cham, Switzerland: Springer, 2014.
- [9] Z. Wang, L. Du, and L. Li, "Visual attention based target detection and discrimination for high-resolution SAR images in complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1855–1872, Apr. 2018.
- [10] D. A. Klein and S. Frintrop, "Center-surround divergence of feature statistics for salient object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2011, pp. 2214–2219.
- [11] L. Duan, C. Wu, J. Miao, and L. Qing, "Visual saliency detection by spatially weighted dissimilarity," in *Proc. Comput. Vis. Pattern Recognit.*, 2011, pp. 473–480.
- [12] Z. Wang, L. Du, and H. Su, "Target detection via bayesian-morphological saliency in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 1–12, Oct. 2017.
- [13] X. Hou and L. Zhang, "Saliency detection: A spectral residual approach," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2007, pp. 1–8.

- [14] J. Zhou, S. Gao, Y. Yan, and Z. Jin, "Saliency detection framework via linear neighbourhood propagation," *IET Image Process.*, vol. 8, pp. 804–814, Dec. 2014.
- [15] J. Ren, X. Gong, L. Yu, W. Zhou, and M. Y. Yang, "Exploiting global priors for RGB-D saliency detection," in *Proc. Comput. Vis. Pattern Recognit. Workshops*, 2015, pp. 25–32.
- [16] J. Wang, H. Lu, X. Li, N. Tong, and W. Liu, "Saliency detection via background and foreground seed selection," *Neurocomputing*, vol. 152, pp. 359–368, 2015.
- [17] Y. Y. Zhang, Z. P. Wang, and X. D. Lv, "Saliency detection via sparse reconstruction errors of covariance descriptors on Riemannian manifolds," *Circuits Syst. Signal Process.*, vol. 35, pp. 4372–4389, 2016.
- [18] H. Lu, X. Li, L. Zhang, R. Xiang, and M. H. Yang, "Dense and sparse reconstruction error based saliency descriptor," *IEEE Trans. Image Process.*, vol. 25, no. 4, p. 1592–1603, Apr. 2016.
- [19] X. Li, H. Lu, L. Zhang, and R. Xiang, "Saliency detection via dense and sparse reconstruction," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2013, pp. 2976–2983.
- [20] W. Qi, J. Han, Y. Zhang, and L. Bai, "Saliency detection via Boolean and foreground in a dynamic Bayesian framework," *Visual Comput.*, vol. 33, pp. 209–220, 2017.
- [21] L. Zhang, M. H. Tong, T. K. Marks, H. Shan, and G. W. Cottrell, "SUN: A Bayesian framework for saliency using natural statistics," *J. Vis.*, vol. 8, pp. 32–1–32–20, 2008.
- [22] Y. Wei, F. Wen, W. Zhu, and J. Sun, "Geodesic saliency using background priors," in *Proc. Eur. Conf. Comput. Vis.*, 2017, pp. 29–42.
- [23] C. Koch and S. Ullman, "Shifts in selective visual attention: towards the underlying neural circuitry," *Human Neurobiol.*, vol. 4, pp. 219–217, 1985.
- [24] Z. Chen *et al.*, "Vehicle detection in high-resolution aerial images via sparse representation and superpixels," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 103–116, Jan. 2016.
- [25] S. S. Gao, C. Jing, L. I. Li, J. B. Zou, and C. M. Zhang, "Saliency detection based on superpixels clustering and stereo disparity," *Appl. Math.*, vol. 31, pp. 68–80, 2016.
- [26] Y. Sun, L. Du, Y. Wang, Y. Wang, and J. Hu, "SAR automatic target recognition based on dictionary learning and joint dynamic sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1777–1781, Dec. 2016.
- [27] J. Wang, J. Yang, K. Yu, F. Lv, T. Huang, and Y. Gong, "Locality-constrained Linear Coding for image classification," in *Proc. Comput. Vis. Pattern Recognit.*, 2010, pp. 3360–3367.
- [28] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, May 2012.
- [29] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognit. Lett.*, vol. 27, pp. 861–874, 2006.
- [30] A. P. Bradley, "The use of the area under the ROC curve in the evaluation of machine learning algorithms," *Pattern Recognit.*, vol. 30, pp. 1145–1159, 1997.
- [31] M. Sciotti, D. Pastina, and P. Lombardo, "Exploiting the polarimetric information for the detection of ship targets in non-homogeneous SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, 2002, pp. 1911–1913, vol. 3.
- [32] Z. Han and J. Chong, "A review of ship detection algorithms in polarimetric SAR images," in *Proc. Int. Conf. Signal Process.*, 2004, pp. 2155–2158, vol. 3.



Lu Li received the B.S. degree in electronic and information engineering from Xidian University, Xi'an, China, in 2015, where she is currently working toward the Ph.D. degree in signal processing with the National Laboratory of Radar Signal Processing.

Her main research interests include synthetic aperture radar automatic target recognition and pattern recognition.

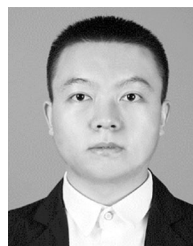


Lan Du (M'11–SM'17) received the B.S., M.S., and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 2001, 2004, and 2007, respectively.

From 2007 to 2009, she was a Postdoctoral Research Associate with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA. She is currently a Professor with the National Laboratory of Radar Signal Processing, Xidian University. Her research interests include statistical signal processing and machine learning with

application to radar target recognition.

Dr. Du was the recipient of the National Excellent Doctoral Dissertation of PR China in 2009 for his doctoral dissertation. Her research work is supported by the National Natural Science Foundation of China for Excellent Young Scholars and Chang Jiang Scholars Program for Young Scholars.



Zhaocheng Wang received the B.S. degree in electronic information science and technology from the Qingdao University of Technology, Qingdao, China, in 2013. He is currently working toward the Ph.D. degree in signal processing with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China.

His research interests include synthetic aperture radar (SAR) automatic target recognition, SAR image processing, and pattern recognition.